

Density Calculation

Definition. Given a physical object occupying a region $\Omega \subset \mathbb{R}^3$ with mass m and volume $V = \int_{\Omega} dV$, the (average) density is defined as

$$\rho = \frac{m}{V}.$$

When mass is distributed non-uniformly, one defines a *local density function* $\rho(x, y, z)$ such that

$$m = \iiint_{\Omega} \rho(x, y, z) dV,$$

recovering the pointwise mass-per-unit-volume in the limit of a vanishing neighborhood around each point.

Worked example. Consider a metal rod of length $L = 2$ m whose linear density varies as $\rho(x) = 3 + x^2$ (in kg/m), for $x \in [0, 2]$. The total mass is

$$m = \int_0^2 (3 + x^2) dx = \left[3x + \frac{x^3}{3} \right]_0^2 = 6 + \frac{8}{3} = \frac{26}{3} \approx 8.67 \text{ kg}.$$

The *average* density over the rod is then $\bar{\rho} = m/L = \frac{26}{3}/2 = \frac{13}{3} \approx 4.33$ kg/m, distinct from the pointwise density at any single location.

Key theorem (Mean Value Theorem for Integrals). If ρ is continuous on $[a, b]$, there exists $c \in [a, b]$ such that

$$\bar{\rho} = \rho(c) = \frac{1}{b-a} \int_a^b \rho(x) dx.$$

This guarantees that the average density equals the actual local density at some point within the domain — a foundational link between integral (global) and pointwise (local) descriptions of mass distribution.

Lab cell (SymPy).

```
import sympy as sp
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x = sp.symbols('x')
rho = 3 + x**2
L = 2

mass = sp.integrate(rho, (x, 0, L))
avg_density = mass / L

print("Total mass:", mass)
print("Average density:", sp.simplify(avg_density))
```